FoodExpert: Portable Intelligent Device for Rapid Screening of Pulse Quality and Adulteration

Harsh Pandey, Subhanshu Arya, Debanjan Das, *Senior Member, IEEE*, and Venkanna Udutalapally, *Senior Member, IEEE*

***Abstract*—Pulses are one of the most important food crops in the world due to their higher protein content, approximately 21%–25%. Therefore, it is crucial to analyze the crop's quality and impurity levels. Stones, pebbles, marble chips, and synthetic dyes, such as lead chromate, metanil yellow, and artificial colors, are some of the impurities added to pulse products, accidentally or on purpose. The existing analysis techniques are mostly laboratory-based, time-consuming, costly, and require human examination. To address this issue, this article presents an intelligent system, FoodExpert, based on image processing that automatically uses an image of a pulse sample to identify the region of interest and essential attributes. Then, machine learning frameworks are used to predict pulse quality and adulteration levels based on the obtained parameters. On the test dataset, the suggested model had a 96% accuracy rate for pulse quality prediction and 94% accuracy for adulteration level prediction. The model was successfully deployed on a Raspberry Pi-based hardware prototype and mobile application.**

***Index Terms*—Adulteration, computer vision, edge computing, food quality, image processing.**

I. Introduction

PULSES are one of the most important food crops for daily protein needs, whose quality is becoming compromised due to uncontrolled adulteration, addition of foreign materials, etc. Consumers usually perform quality checking at the individual level. After cooking, consumers test the pulse quality based on their color, odor, and taste. In addition, they also look for broken seed grains and other foreign particles at the time of purchase. In the industries, the test is performed based on cooking time, protein percentage, water absorption, and pulse recovery [1]. However, there is another perspective for quality checking, that is, for the farmers who produce them. They lack the technology to test their products, leading to their sale at an inappropriate price. Furthermore, pulse adulteration significantly impacts human health; for instance, it can cause stomach disorders and occasionally even cancer. In particular, economically inexpensive adulterants could have detrimental effects on public health. In India and many other countries, pulses are often adulterated with dyes, such as metanil yellow, lead chromate, and synthetic colors, to make them appear brighter. These adulterants are carcinogenic [2] and often cause digestive problems. Therefore, screening the pulse quality for adulteration detection is becoming an integral part of food safety protocols.

The existing methods for detecting adulterants in pulses are based on either physical or chemical processes [3], [4], [5]. However, these techniques are destructive, manual, and tedious in process. To speed up this process, image processing has been employed in recent studies on rice quality checking, such as [6] and [7]. However, they require additional hardware setup and calibration to reduce the portability and accessibility of the system. Moreover, Lanjewar et al. [2] offered a way to identify tartrazine-colored rice flour adulteration in turmeric using image processing and a convolutional neural network (CNN), but this method requires continuous Internet access for proper functioning. In another work, Mandal et al. [8] used image processing and machine learning (ML) to identify adulterants in turmeric powder but failed to illustrate the levels of adulteration. Food adulteration levels can also be verified using DNA-based methods, as shown in [9] and [10], where DNA microsatellite analysis was utilized to detect adulterated wheat. Knödler et al. [11] suggested a method to gauge the authenticity of wheat flour and pasta by examining alkylresorcinol composition. However, these techniques require laboratories and other industrial settings, challenging their use and location.

To address the aforementioned limitations, we propose FoodExpert, a hardware device that can be installed in grocery stores and government-based food authorities to check pulse quality rapidly in a nondestructive manner. The proposed method employs image processing combined with ML to automatically segment the region of interest (ROI) and extract features for categorization. The system includes a camera that serves as a sensor, capturing an image of the pulse sample, determining the dimensional and color attributes, and estimating the quality and metanil yellow adulteration level of the pulse sample.

The major contributions of this work are as follows.

1) This article proposes an automated, handheld device with an associated mobile app-based solution, FoodExpert, that

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can use images collected by a camera to identify the quality grade and adulteration of pulse samples.

2) The proposed model uses color space features to distinguish between various adulteration levels efficiently and uses dimensional properties, including eccentricity, area, cross-axis ratio, etc., to compute pulse grade on-board.
3) The proposed FoodExpert solution determined pulse quality with 94% accuracy. The model was also tested by real-life deployment with an accuracy of 87%. Random grain color features were approximated with 0.096 RMSE in the photos, and their dimensions were computed with an error margin of $\pm$ 0.02 cm.
4) The portable device can be operated seamlessly under typical marketplace lighting conditions, helping shopkeepers, consumers, and government authorities to check the quality of pulses and make informed decisions. The solution offers a more sophisticated approach by providing users with a comprehensive index, on a scale of 0–100, of the adulteration level and pulse quality.

II. Related Works and Research Gaps

A. Food Grain Quality Measuring

The estimation of pulse quality can be characterized as manual or computer-based. Many commercial procedures involve manual measurement of the length and width of multiple grains using Vernier calipers and grinding the pulse to flour to understand various chemical aspects and assess its grade. Invasive methods, such as [1], that use physical techniques to extract features, such as protein (%), pulse recovery (%), water absorption (%), and cooking time, require users to judge the quality using the extracted features. Recently, several computer vision-based methods have been developed to assess the quality of wheat grain [12], [13], [14]. The offered solutions are based on ML in most situations [12], [13]. Although much work has yet to be based on deep learning, there are CNN-based algorithms for classifying wheat [15], where feature extraction was performed using CNN on rice grain samples. Kar et al. [16] proposed annotation-free quality estimation of wheat grains using deep neural networks. Similarly, the authors in [17] and [18] presented computer vision and deep learning-based methods for classifying coffee beans.

B. Food Grain Adulteration Detection

Traditional approaches to detect adulterants in various food items, such as [4], [5], and [19], are based on either physical or chemical processes. Nag et al. [4] described the development of a nickel–cobalt oxide nanoparticle-modified graphite paste electrode ($NiCo_2O_4$@GP) for detecting and quantifying metanil yellow traces in food particles. Chakraborty and Biswas [5] proposed a method to detect milk adulterants, such as ammonium sulfate, sodium hydroxide, sodium-bi-carbonate, and common salt, using an impedance sensor that is sensitive to the ionic concentration of the measuring medium and gives different phase angles of the impedance when immersed in different adulterated/unadulterated milk. Further, Ray et al. [19] proposed a paper-based microfluidics-based technology for detecting specific adulterants, followed by adding samples to obtain well-defined qualitative or quantitative signals to confirm the presence of the target species. Food adulteration levels can also be verified using DNA-based methods [9] [10], where DNA microsatellites are analyzed to detect adulterated wheat. Knödler et al. [11] suggested gauging the authenticity of wheat flour and pasta by examining alkylresorcinol composition. But, these techniques need labs and other industrial settings, making them challenging to use and locate.

Further, Teye and Amuah [20] focused on detecting rice varietal integrity and adulteration fraud using chemometric analysis of pocket-sized near-infrared (NIR) spectra data. However, reliance on pocket-sized NIR spectra data might introduce variability due to device limitations and external factors. Bai et al. [21] introduced a novel approach to swiftly and nondestructively identify sorghum adulteration using optimization algorithms and hyperspectral imaging techniques. However, there is a need for complex optimization algorithms that might require significant computational resources and the challenge of ensuring the accuracy and reliability of hyperspectral imaging in real-world scenarios. Yang et al. [22] presented a machine vision-based approach for identifying adulteration in granular raw materials during Baijiu brewing, showcasing its potential to maintain the quality of the brewing process. Its limitations in handling variations in granular material textures, sizes, and lighting conditions could impact the accuracy and robustness of the machine vision system.

C. Implementation

Recent efforts on rice, such as [6], [7], and [23], have incorporated image processing to speed up the procedure. However, they require additional hardware setup and calibration, reducing the system's mobility and accessibility. Some methods, such as [3], [15], and [24], use image processing with image segmentation (K-means algorithm) and applying classifiers, such as support vector machine (SVM) and K-nearest neighbor (KNN) for food inspection. In addition, Parmar et al. [25] combined image processing with ML to develop a separate application for analyzing the quality of various types of food. Despite this, it has not yet been deployed on smartphones. They also demand additional equipment, which limits their general use. In [26], the hyperspectral imaging technique has been used for data creation and extraction; then, various ML algorithms are used for seed quality and safety inspection. Similar methods are presented in [27] and [28] by merging hyperspectral imaging with several ML algorithms.

D. Shortcomings of Existing Works

The research gaps in the existing methods as summarized in Table I are as follows.

1) The user needs to judge the final step in manual solutions, which leads to vague judgment.
2) Few images processing-based solutions require manual quality inference from the extracted parameters.

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TABLE I
COMPARISON BETWEEN EXISTING RESEARCH

References	Image Type	Image Background	Segmentation	Classification	Application
[15]	RGB	Black	Canny	CNN (customized architecture)	Rice Grain purity analysis
[17]	HSV/ CIELAB	White	Thresholding	K-NN	Quality coffee beans
[18]	RGB	Black	Grayscale thresholding color detector	CNN (customized architecture)	Grain quality detection
[28]	Hyperspectral	Black	Thresholding	CNN (customized architecture)	Identify rice seed varieties
[14]	RGB	-	K-means clustering	SVM	Wheat grain classification
FoodExpert (proposed)	**RGB**	**Black**	**OTSU+ Modified Canny**	**Random Forest Classifier**	**Grading Pulse Quality**

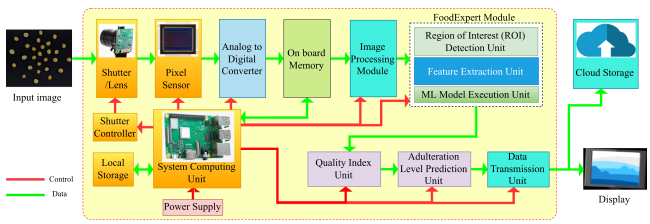
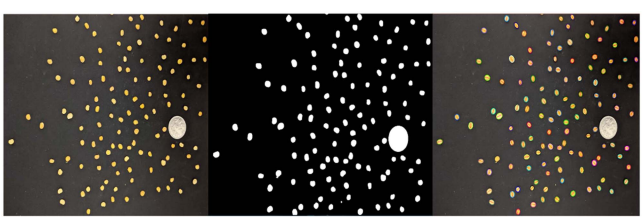


Fig. 1. Proposed FoodExpert architecture.



(a) (b) (c)

Fig. 2. ROI segmentation and feature extraction. (a) Original image. (b) Result of binarization and morphological processing. (c) Final extracted ROIs after contour filtering.

3) Automatic image processing systems necessitate fixed equipment, which may produce mistakes due to slightly shifted equipment.
4) Prior solutions are not scalable and portable for easy dissemination to remote places as they require expensive setup and experienced staff.
5) There is no specific work in the field of pulses (lentils).

III. FOODEXPERT: AN OVERVIEW OF A QUALITY EVALUATION TOOL ON A FUNDAMENTAL LEVEL

This section explains the FoodExpert module's three stages for determining pulse grain quality. Fig. 1 shows the hardware architecture of the FoodExpert system, which is explained in detail in the following sections.

A. Stage-1: ROI and Features Extraction

ROI detection unit of FoodExpert (see Fig. 1) segments the ROI from the captured image. First, an image of the pulse grains is collected as an input, assuming that the grains are spaced out for the best outcomes [as shown in Fig. 2(a)]. The various ordered functions applied on the input image are discussed below.

1) Gaussian Blurring: Gaussian blurring reduces image noise and detail. The visual effect of this blurring technique is a smooth blur that resembles that of viewing an image through a translucent screen. First, the input image was converted to grayscale and passed through a Gaussian filter with a kernel size 7×7 to smoothen and denoise the image, the choice of 7×7 kernel size and a sigma value of 1.2 is often used as a tradeoff between smoothing and preserving edges. The larger the kernel size, the more smoothing will occur, but also, more edges will be blurred. Equation 1

$$G(x, y, \sigma) = \frac{1}{2 \times \pi \times \sigma^2} \times e^{-\frac{x^2+y^2}{2\times\sigma^2}} \tag{1}$$

determines the amount of change that needs to be applied to each pixel where (x, y) denotes the pixel coordinates and σ represents the variance value in the neighborhood. Next, Otsu's thresholding, named after Nobuyuki Otsu, is performed to binarize the image and then pass it for further morphological operations.

2) Morphological Operations and Contour Extraction: A set of morphological operations, i.e., erosion and dilation are then performed on the binarized picture using a rectangular structuring element of size 7×7. The structural element glides over the whole image during erosion, and a pixel value is set to 1 only if all of the pixel values under the kernel are also set to 1; otherwise, it is set to 0. During dilation, however, a pixel value is set to 1 if at least one-pixel value under the kernel is set to 1. In other words, erosion thins the limits of foreground items, whereas dilation thickens the boundaries of foreground objects. These techniques assist in eliminating picture noise and give the pulse grains a distinct form by separating neighboring grains. Next, contour detection is performed on the resultant image using the modified Canny edge detection [29] with a σ value of 0.33, this returns a list of points enclosing the outlines of grains by detecting a change in pixel values. Generally, a larger value of σ will result in thicker edges, whereas a smaller value will produce thinner edges. Hence, we used a sigma $= 0.33$ on our image, which usually fits well on images that have already gone through erosion and dilation. These points comprise the ROI [see Fig. 2(b)] utilized for further processing.

3) Determining the Conversion Factor and Filtering the Contours: The pulse quality is determined by the pulse grains' length, thickness, and eccentricity. All of these characteristics are calculated in pixels. When using a camera to capture photographs, the distance between the pulse and the camera cannot be guaranteed, which might result in different diameters of pulse grains in several images of the same sample because of the number of pixels in a region of the image varies with distance from the item. A conversion factor is calculated using a reference item in the image to compensate for this. The reference object for this investigation is a coin with a known radius of 25 mm. The identified contours are sorted by area, with the largest being considered. This contour is fitted with a rectangular box of the smallest area, with the vertices having the slightest average mean square error. The length and breadth of the box are returned by the OpenCV version of box fitting (2), whose average is taken to ensure that we acquire the diameter of the coin appropriately

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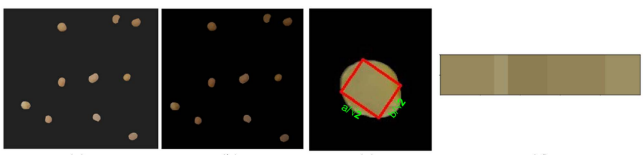
Algorithm 1: Filter_Contours.

Input: contours
Output: final_contours
final_contours $\leftarrow \varnothing$ {Initialize array as Null}
Estimate **contour_areas** from **contours**
mean_contour_area $\leftarrow \frac{\Sigma \text{contour_areas}}{\text{Number of contours}}$
for each **contour_area** in **contour_areas do**
  if contour_area > 0.5(mean_contour_area) **and contour_area** < 2.0(mean_contour_area) **then**
    final_contours $\leftarrow$ **final_contours** $\cup$ **contour_area**
  end if
end for

Algorithm 2: Extract_Features.

Input: PPMM and contours
Output: physical_features
final_contours = **Filter_Contours**(contours)
for each **contour** in **final_contours do**
  if contour has at least 5 points **then**
    a = max(d1, d2), **b** = min(d1, d2)
    elliptical_area $= \pi \times a \times b$
    contour_area = find area of contour
  end if
end for
physical_features $= \frac{\sum_{i=1}^{N} \text{Feature}}{N}$, N = Number of contours



(a) (b) (c) (d)

Fig. 3. (a) Original image. (b) Result of masking over each contour. (c) Maximum area rectangle fitted. (d) Histogram bar of dominant colors in rectangular portion using K-means clustering.

(3). The output is divided with 25 mm to get the Pixels Per MilliMeter (PPMM) value (4):

$$\text{Center, Width, Height} = \text{cv2.minAreaRect(coin)} \tag{2}$$

$$\text{radius_pix} = \frac{(\text{Width} + \text{Height})}{2} \tag{3}$$

$$\text{PPMM} = \frac{\text{radius_pix}}{25}. \tag{4}$$

In general, noise in the image affects contour detection, which detects outlines of tiny white blobs in the image, which can substantially impact the assessment results. To avoid this, the grains are arranged so that they do not come in contact with one another, and the contours are filtered by simply rejecting the extremely small or large ones. This filtering of contours is performed using *Filter _ Contours* illustrated in the Algorithm 1. First, compute the mean of all the contour regions (except the coin's area). Then, higher and lower threshold values are estimated in the mean area, and any contour whose area does not fall within them is disregarded. Experimentally, the lower and upper thresholds are obtained as 0.5 mean area and 2.0 $\times$ mean area. The final contours are the required areas in the image or the ROIs. These areas contain pulse kernels, which are used for analysis.

4) Dimensional Feature Extraction: The following physical characteristics are taken into account by FoodExpert: Major axis, minor axis, MinbyMaj ratio, eccentricity, and grain area. We have found that the model produces the best recommendations throughout the analysis when these features are included. An approximation of the contour's shape aids us in swiftly determining the contour's dimensions. Because pulse grains are roughly circular, we used least squares to fit an ellipse to the contour and establish its boundary. The obtained length and breadth of the rectangular region are converted to actual-world dimensions using the PPMM conversion factor determined previously (4)

$$d_1, d_2 = \text{cv2.fitellipse} \tag{5}$$

$$a = \max(d_1, d_2),\ b = \min(d_1, d_2) \tag{6}$$

$$\text{elliptical area} = \pi \times a \times b. \tag{7}$$

The major and minor axes are represented by a and b, respectively. Equation (6) calculates the major and minor axis lengths. Equation (7) is then used to determine the approximated elliptical area of the pulse grain, followed by eccentricity and MinbyMaj ratio calculation. MinbyMaj ratio is the ratio of the minor and major axes of the pulse grain. These describe the amount of elongation present in the pulse grain and are generally used to identify the pulse's shape and type, Fig. 2(c) illustrates the final ROIs extracted after contour filtering and feature extraction. The *Extract _ Features* algorithm (see Algorithm 2) depicts the steps involved in extracting the features using a single loop over the obtained ROIs.

5) Color Space Adulteration Detection: After contour filtering, the backdrop of the resultant image is not entirely black, as shown in Fig. 3(a), which can add stray colors to the ROI color space. Therefore, each contour is masked and overlapped on a pure black background, as shown in Fig. 3(b). However, some grain contours may still include the background pixels, which can cause a wrong selection of color space information. In this endeavor, a rectangle of the maximum area is fitted inside the contour to filter out the unwanted background pixels, as depicted in Fig. 3(c).

Then, K-means clustering is used to extract color values, with the number of bins set at three. It gives color bins [see Fig. 3(d)] that predominate in each rectangle-shaped contour area, which is used as a feature vector for training.

6) Pulse Index Formulation: The grade and adulteration level models classify a sample image into their respective classes. Further, our prototype provides an index for both models to quantify the predictions further. The pulse quality index is calculated based on (8), where G_1, G_2, and G_3 are the percentage of pulses in respective grades in a sample image. The coefficients of G_1, G_2, and G_3 are weights of a particular grade, which are decided based on the physical dimensions of a grade with respect

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to the best-quality pulse. Equation (8) provides a scale between 50 and 100, where 100 indicates the best-quality pulse

$$Q_{\text{index}} = 1 \times G_1 + 0.7 \times G_2 + 0.5 \times G_3. \tag{8}$$

Like the above formulation, the adulteration level index is represented by (9). AL_1, AL_2, and AL_3 represent the percentage of pulses in the corresponding adulteration levels, and the concentration of adulteration levels determines their respective coefficients. This equation will provide an index ranging from 0 to 100, with a higher number indicating more adulteration

$$A_{\text{index}} = 0 \times AL_1 + 0.5 \times AL_2 + 1 \times AL_3. \tag{9}$$

B. Stage-2: Dataset Preparation and Model Training

For two different features extracted, we propose two different models running independently to output the quality of the given sample and the amount of adulteration present. We have generated the dataset with two pulses, namely Pigeon peas and Chickpeas, which closely resemble the method used for their adulteration, i.e., metanil yellow. The images of the prepared sample pulses are captured using a 12-megapixel (Pixel-3) and a 64-megapixel (POCO-X3) camera system. The dataset was prepared under controlled illumination conditions with an intensity of 400 lux and a color temperature of 4400 K using a white light source with a split of 70-20-10 for training, validating, and testing, respectively.

1) Pulse Quality Dataset: We created a balanced dataset of 130 photos for the pulse grading model, where the dataset was divided into three grades. Grade 1 consists of 40 images, Grade 2 consists of 45 images, and Grade 3 consists of 45 images, with more than 20 data points in each of the images, which constitute our dataset for pulse quality. Given the inherent limitation of captured photos, we adopted a per-grain basis for data collection. This approach, capturing individual grains within each image, substantially augmented the dataset size, allowing us to address data scarcity effectively.

1) Grade 1—No broken pieces of a pulse.
2) Grade 2—50%–75% of broken pieces in pulse sample.
3) Grade 3—more than 75% of broken pieces in pulse sample.

All previous procedures are merged to extract the characteristics from each dataset image. These options are saved in a comma separated values (CSV) file. The CSV files of various grade grains are combined, and a random forest classifier is trained to classify pulse samples based on these characteristics. A random forest is a meta-estimator that fits several decision tree classifiers on various subsamples of the dataset and uses averaging to improve the predictive accuracy and control overfitting. During training, the decision tree develops node-like structures in which the decision rules are written. The estimator uses these rules to create predictions (see Fig. 4).

2) Adulteration Dataset: A balanced dataset of 120 photos was constructed for this model similar to the pulse grade dataset, we adopted a per-grain basis for data extraction. Each adulteration level has 40 images with more than 20 data points in each of the images, which constitute our adulteration dataset. We created three batches of the dataset for adulteration level estimation

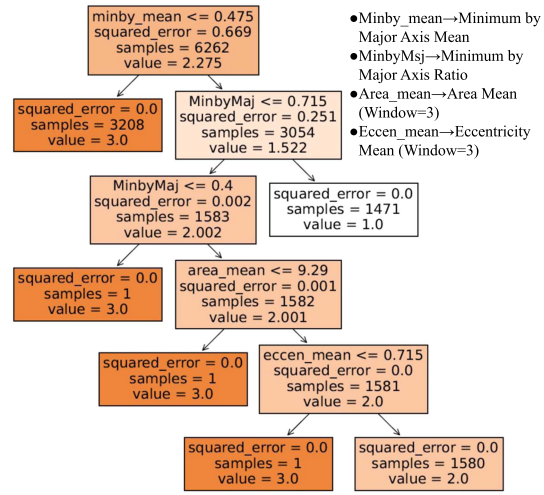

Fig. 4. Visualization of an arbitrarily picked decision tree of the random forest.

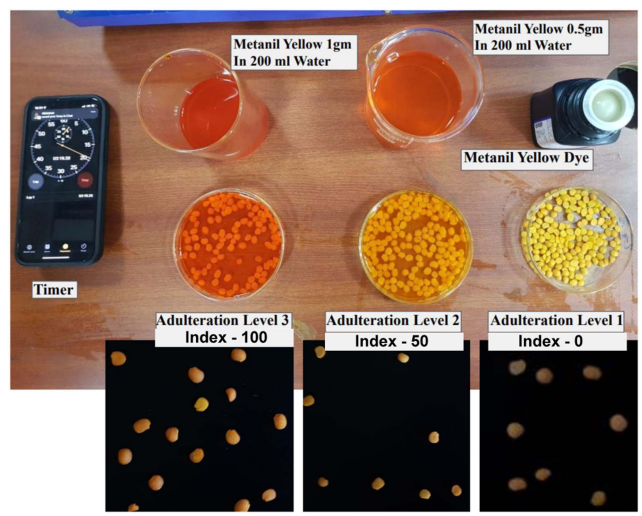

Fig. 5. Dataset preparation for different levels of adulteration using metanil yellow.

(see Fig. 5), where index 100 refers to maximum adulteration as per the adulteration index and 0 index vice versa.

1) Adulteration level 1—No adulteration.
2) Adulteration level 2—Adulterated by soaking in a solution of 0.5 gm metanil yellow dye and 200 ml water, then heated for 10 min at 60° to remove excess water.
3) Adulteration level 3—Adulterated by soaking in a solution of 1 gm metanil yellow dye and 200 ml water, then heated for 10 min at 60° to remove excess water.

These features are saved in a CSV file. The CSV files of varied adulterant levels are combined, and next, a random forest classifier is trained to classify the pulse samples based on adulteration levels.

IV. Field Testing and Deployment

A. Performance of the ROI Segmentation

We developed an efficient ROI segmentation technique that successfully extracted 98% of the area of interest. Our ROI segmenter uses image processing techniques and ML algorithms

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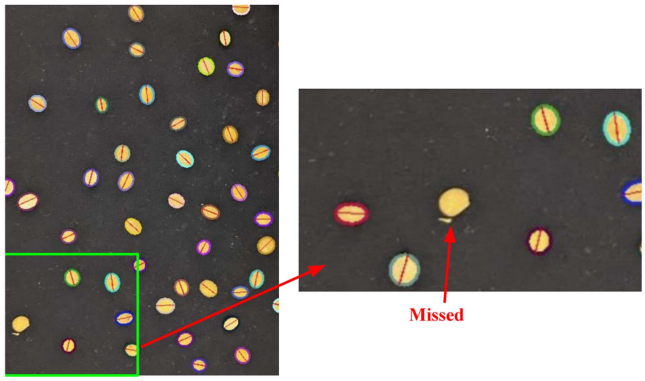
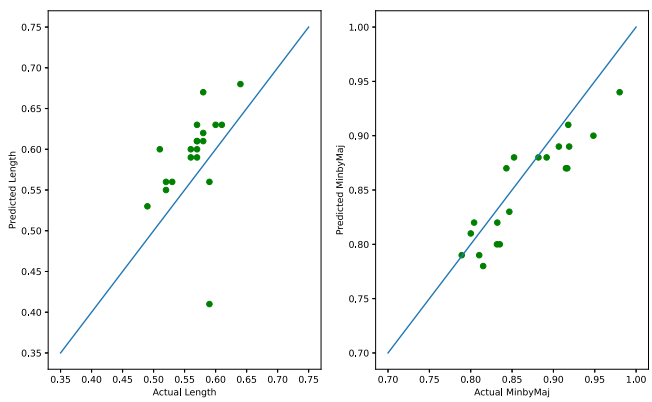


Fig. 6. Extraction of ROI with highlighted contour



(a) (b)

Fig. 7. High correlations between (a) extracted and actual length and (b) extracted and actual MinbyMaj ratio features of pulse grains.

to accurately identify and extract desired regions. This approach allowed us to achieve a high level of accuracy, with 98% of the area of interest being successfully extracted. The system correctly detects the pulse grains in the test image for any mixture of pulse grains, including full-length, broken, or damaged ones, as shown in Fig. 6. In addition, the ROI segmenter's efficiency is noteworthy, as it can process images quickly with minimal computational resources.

The following section will discuss the sample results for pulse quality and adulteration level. The results for each section have been shown separately, highlighting the key metrics and performance of the models. This section aims to analyze the two models and their performance metrics comprehensively.

B. Performance and Error Analysis of Pulse Quality Index

1) Extraction of the Physical Features: We examined the findings predicted by FoodExpert across a few grains of the pulse at random to assess the accuracy of the proposed model. The correlations between the extracted and actual lengths and the MinbyMaj ratio are shown in Fig. 7.

Dimensions with an RMSE of $\pm$ 0.365 mm and a 5% mean absolute percentage error are computed from 30 grains randomly selected from three photos. The grains were manually measured using a 0.01 mm precision measuring scale, meaning the extracted characteristics might be more accurate than those

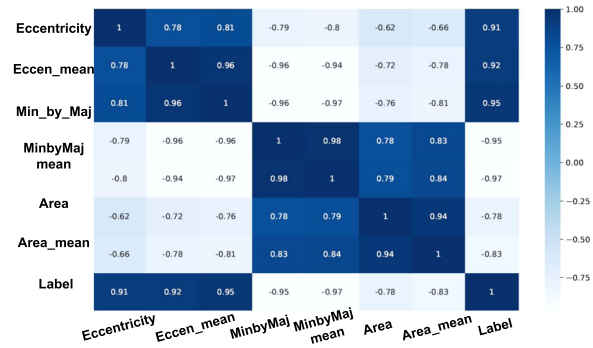


Fig. 8. Importance of each physical feature plotted using a heat map.

TABLE II
QUALITY CLASSIFICATION METRICS

Grade	Precision	Recall	Specificity
1	0.94	1.00	1.00
2	0.85	0.88	0.88
3	0.93	0.78	0.77

physically measured. This proves that the FoodExpert model can precisely predict the physical characteristics of most pulse grains in terms of real-world dimensions, demonstrating its accuracy and self-sufficiency.

Fig. 8 describes each feature's importance while assessing the pulse sample's quality. As expected, the eccentricity of the pulse grains is highly correlated with the predicted pulse quality. The importance of features is calculated based on the correlation value with the pulse grade.

2) Prediction of Pulse Sample Quality: We used standard metrics viz., recall, precision, F1-score, and accuracy in assessing FoodExperts's prediction efficacy. The averages of all extracted characteristics for each pulse grain in an image are calculated, and a set of features that describes the complete image is formed. The traits were predicted autonomously with excellent accuracy, as described above. On the test set, the model was evaluated by plotting a confusion matrix and producing a classification report.

Table II gives the classification report created from the confusion matrix and other metrics, including precision, recall, and specificity, which are used to evaluate the FoodExpert model's performance. Because 1-star and 3-star ratings have about equal numbers of good and bad grains in each sample, their precision, recall, and specificity are higher than other ratings. Few 2-star and 3-star samples contain an overlapping set of features, lowering the metrics' values and, in some situations, causing a challenge for the classifier.

3) Statistical Tests: The Cohen-kappa statistical test is further used to assess the robustness of our model. The Cohen-kappa score, also known as the Cohen-kappa coefficient, measures the agreement between the actual and expected outcomes while taking a chance into account [30]. A score of 0.8553 denotes the proposed model's reliability. As illustrated in Fig. 9, we plotted confusion matrices between the actual and anticipated outcomes of the test dataset to support our assertion further. The

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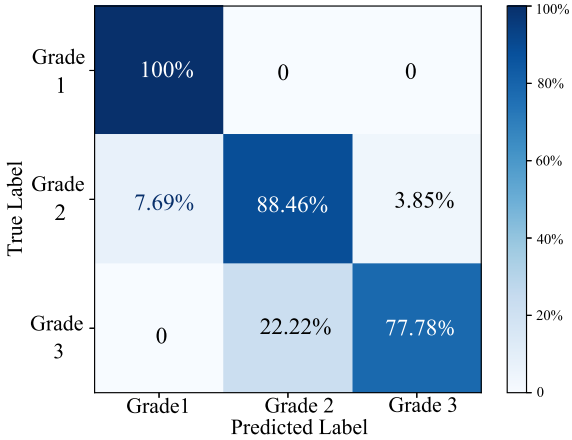


Fig. 9. Confusion matrices for dimension dataset.

TABLE III
COMPARISON BETWEEN MODELS

Ranking	Model	SD	Val
1	Random Forest Classifier	0.195	0.9903
2	Logistic regression Classifier	0.198	0.9817
3	XGBoost Classifier	0.201	0.9813
4	Elastic Net Classifier	0.234	0.9764
5	LightGBM Classifier	0.277	0.9612

SD: Standard Deviation and Val: Validation score.

confusion matrix describes the similarity between the actual and expected values of the data presented in our model. As shown in Fig. 9, FoodExpert correctly quantifies the majority of the grades for the pulse sample, The relatively low accuracy associated with the true label of 3 revealed a key challenge originating from the high-dimensional resemblance between grades 2 and 3. The nuanced similarities between these two grades present a significant hurdle in accurate differentiation, even for complex models.

C. Performance and Error Analysis of Pulse Adulteration

1) Accurate Model Selection: In our work, we have used EvalML to automate the selection and tuning of ML models. By utilizing this library, we create a model performance ranking based on various evaluation metrics. Once the evaluation metrics have been calculated, the library generates a performance report that lists the models in order of their performance. From the performance report, we can easily identify the best-performing model based on the evaluation metrics of interest. This best-performing model can then be selected for deployment in a production environment. It is important to note that the evaluation metrics used should be selected based on the specific problem and use case at hand. Table III summarizes the performance of the tested models and shows that the random forest classifier has the highest accuracy of 99.03%.

2) Prediction of Pulse Adulteration: The performance of the random forest classifier is evaluated with the common metrics viz., recall, precision, F1-score, and accuracy [31]. On the test set, the model was evaluated, producing a classification report (see Table IV), which reports an accuracy of 91%.

TABLE IV
ADULTERATION CLASSIFICATION METRICS

Adulteration Level	Precision	Recall	Specificity
1	0.98	0.95	0.98
2	0.78	0.83	0.92
3	0.89	0.88	0.93

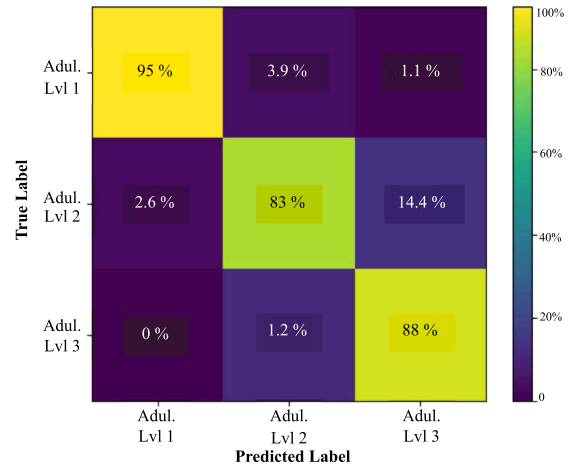
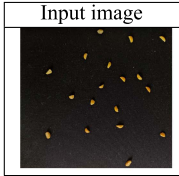
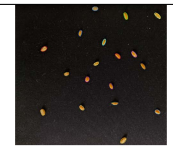
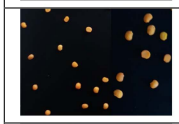
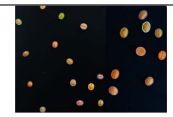
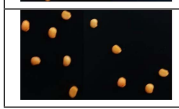
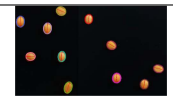


Fig. 10. Confusion matrices for color dataset.

TABLE V
RESULTS OF PREDICTED A_{INDEX}

Input image	ROI	Predicted	Actual
		56	61
		14	11
		64	72
		72	68

This model has a Cohen-kappa score of 0.864430, greater than the standard value of 0.8. This high rating denotes the proposed model's reliability, which can be further improved with additional data points. As illustrated in Fig. 10, we plotted confusion matrices between the actual and anticipated outcomes of the test dataset to support our assertion further. The confusion matrix describes the similarity between the actual and expected values of the data presented in our model. As shown in Fig. 10, FoodExpert properly quantified the majority of the classes for the pulse sample.

3) Quantitative Result of Adulteration Index: The quantitative results of the adulteration index are given in Table V. This

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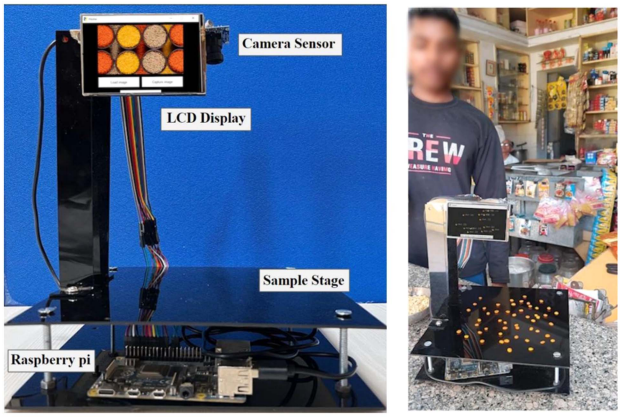

(a) (b)

Fig. 11. (a) Hardware prototype. (b) System being used in the grocery store.

TABLE VI
CAMERA SPECIFICATIONS FOR REAL-LIFE TESTING

Camera Used	Mega Pixels	Aperture	Focal Length	Sensor Size
RPi Camera	5	f/1.8	3.6mm	1/4.00"
OnePlus 9 Pro	8	f/2.4	77 mm	1/2.41"
Pixel 3	12	f/1.8	28 mm	1/2.55"
Nothing Phone 1	16	f/2.45	24mm	1/3.1"
Poco X3 (Scaled down)	24	f/1.89	26 mm	1/1.73"
Samsung Galaxy A54	32	f/2.2	26mm	1/2.8"
OnePlus 10 Pro	48	f/1.8	23mm	1/1.43"
Poco X3	64	f/1.89	26 mm	1/1.73"

table presents the input image, ROI image, predicted adulteration index(A_{index}), and actual adulteration index(A_{index}) for four samples analyzed. Here, A_{index} is calculated using (9).

D. Implementation and Efficiency

A hardware prototype and smartphone implementation are developed based on the proposed architecture shown in Fig. 1. The hardware prototype, as shown in Fig. 11, comprises an liquid crystal display (LCD) screen, a low-power camera sensor, acrylic-made pillars, a sample stage, and a Raspberry Pi (RPi) with 4 GB of RAM and kernel version 5.15 running at a clock speed of 1.5 GHz. The trained model random forest classification was deployed on an RPi, which uses at most 85%–90% of its CPU's capacity during output generation and around 30%–45% during image onboarding.

Further, to validate the system to run in real time, we calculated the power consumption of the RPi. The R Pi consumes 4.5 W in standby mode, whereas it is 6 W during the execution of the model. Furthermore, to assess the real-time accuracy of the prototype, it was tested in an open environment, such as in a retail market, as shown in Fig. 11(b). The result shows the same accuracy as in the case of a laboratory environment.

We developed the FoodExpert application (see Fig. 12) on Android smartphones, as mentioned in Table VI. The correlations between the physical properties evaluated by FoodExpert and the features measured manually are vital for all devices, as shown in Fig. 7. Our model provides nearly the same levels of accuracy of the application's predictions across various devices with various camera kinds. Fig. 13 compares the prototype's performance on different cameras with varying megapixels. The smartphone

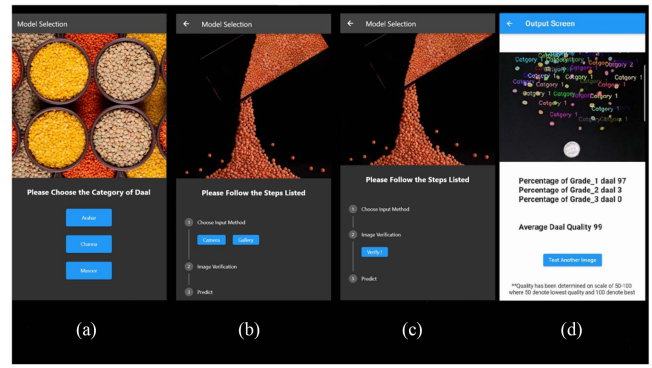

Fig. 12. Different screenshots of the FoodExpert App. (a) Choosing the type of pulse. (b) Choosing gallery or camera. (c) Verifying the captured for prediction. (d) Final result prediction.

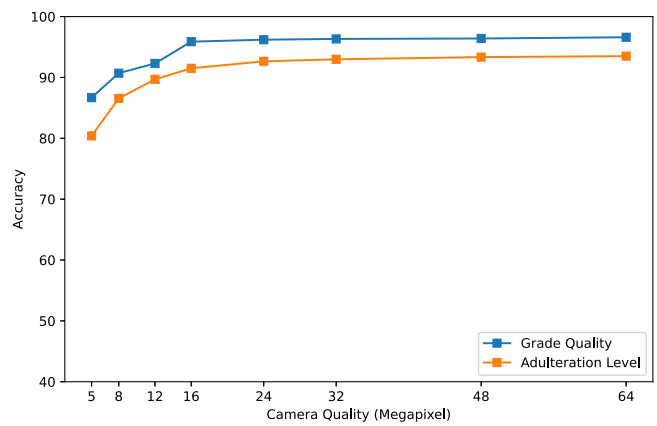

Fig. 13. Performance comparison of the prototype on different cameras with varying megapixels.

cameras with different characteristics, as mentioned in Table VI, are used for testing the FoodExpert performance post deployment. This supports the premise that the software works well on various devices, allowing us to realize our aim of developing a low-cost smartphone model that is widely available.

Regarding the implementation on a smartphone, our system was developed as an Android application using Dart, OpenCV and Chaquopy (for running Python scripts in Dart) library. We tested the system on a smartphone with 4 GB RAM running Android OS. We measured the CPU and memory usage of the system using the Android Profiler tool. Our experiments showed that our system had minimal computational requirements on the smartphone and was able to run smoothly. Typically, grading of daal took about 3–4 s on this smartphone. We conducted a profiling analysis of our classifier using the cProfile module in Python to provide a performance breakdown. The feature extraction process was our classifier's most computationally expensive component, accounting for approximately 80% of the total runtime. The remaining 20% of the runtime was spent on the actual classification step. Overall, we believe that our implementation of the random forest classifier is efficient and effective, with most of the runtime spent on the feature extraction process.

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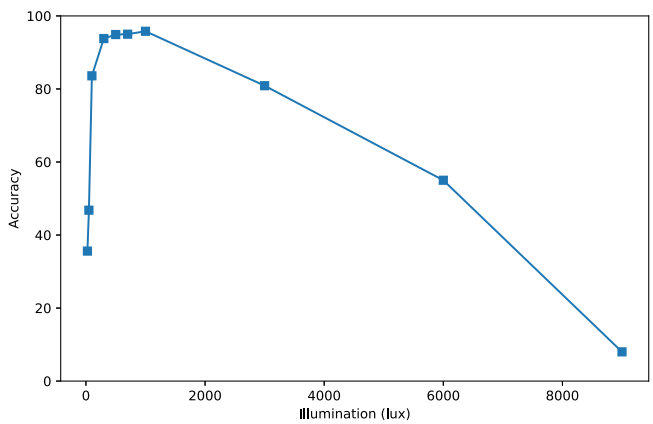

Fig. 14. Pulse grade model performance on varied light condition.

E. Optimum Lighting Condition

The lighting conditions used during the training and testing of our proposed method are critical for achieving accurate and reliable results. Lighting conditions can affect the color and texture of the pulses, which can impact our method's accuracy in detecting adulteration levels. When the surface is smooth, black, and nonreflective, FoodExpert produces the best results. We conducted additional experiments to evaluate the effect of lighting conditions on the accuracy of our proposed system. Specifically, we evaluated the pulse quality model's performance under different lighting conditions, including natural, artificial, and dim, and attached the quantitative results in Fig. 14. The results showed that the lighting conditions did not significantly affect the accuracy of the pulse quality model in the 100–1000 luminous intensity range (lux).

A nonreflective paper has been attached to the sample stage area to reduce the effect of the reflective surface. We believe that the optimum lighting conditions identified in our study can serve as a helpful starting point for end-users to optimize the lighting conditions for using our proposed method and improve the accuracy and reliability of the method in real-world scenarios. However, the light temperature (color) range is paramount due to its significant impact on color space analyses, especially in the case of the adulteration model. The temperature of the light source directly influences the color characteristics of the captured images. We conducted our experiments under controlled light temperature conditions to ensure accurate and consistent results. Our investigations revealed that the optimal light temperature for our system is within the range of 4000–5000 K. This choice of optimal light temperature enables the system to effectively capture and process color information, thereby enhancing the accuracy of the adulteration model. It is important to note that while the adulteration model is sensitive to changes in light temperature, the pulse quality model remains relatively stable.

V. Conclusion

This article proposes FoodExpert, an automated, handheld device with an associated mobile app-based solution that can use images collected by a camera to identify the quality grade and adulteration of pulse samples, all within a unified platform. The major conclusions of this article are listed as follows.

1) The proposed method presents an effective and reliable approach for accurately detecting and grading pulse quality, achieving an accuracy of 94% by leveraging contour detection and ML techniques.
2) The proposed method offers a robust approach for detecting adulteration in pulse samples by utilizing a simple color space extracting algorithm, achieving a 91% accuracy in identifying metanil yellow as an adulterant.
3) The proposed method has been implemented in both a portable handheld device and mobile application for onsite assessments.

Thus, it can be confirmed that our system is accurate and efficient enough to revolutionize the food industry by offering a rapid, precise, and automated solution for assessing pulse quality and detecting adulteration. However, there are some areas of improvement for the proposed model. For example, the model can be optimized for variable image lighting conditions to ensure consistent performance across diverse luminous intensities and color temperatures. Moreover, incorporating image warping into the system can lead to more precise measurements of pulse grain dimension. Furthermore, expanding the system's capabilities to detect common adulterants beyond metanil yellow will enhance its versatility and applicability.

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